Graphs and Their Relation to d-Edge Density

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Introduction

Our summer has been focused on Graph Theory, mainly dealing with the combinatorial object consisting of edges and vertices.

A **finite (simple) graph** is an ordered pair $G = (V, E)$ where:

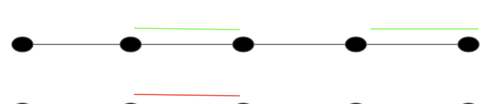
- $V = \{v_1, \ldots, v_n\}$ is a finite set of **vertices**,
- E is a finite set of **edges**, each being an unordered pair of distinct vertices from v. Put more rigorously:

$$E \subseteq \{\{v_i, v_j\} \mid v_i, v_j \in V, i \neq j\}.$$

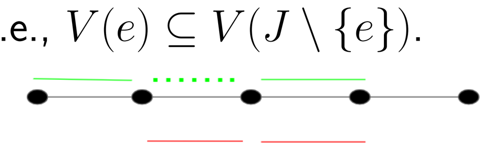
Definitions

These are important definitions for understanding edge density.

- **Edge Blanket**: Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. A subset of edges $J \subseteq E(G)$ is an *edge blanket* if for all $e \in E(G)$, we have $V(e) \cap V(J) \neq \varnothing$.



- **Internal Edges**: A subset of edges $J \subseteq E(G)$ has an *internal edge* if there exists an $e \in J$ that is internal, i.e., $V(e) \subseteq V(J \setminus \{e\})$.



- **d-Edge Density**: Let d be a positive integer. A graph G is *d-edge dense*, if for all subsets $J \subseteq E(G)$ of size d, J is an edge blanket or it has an internal edge.

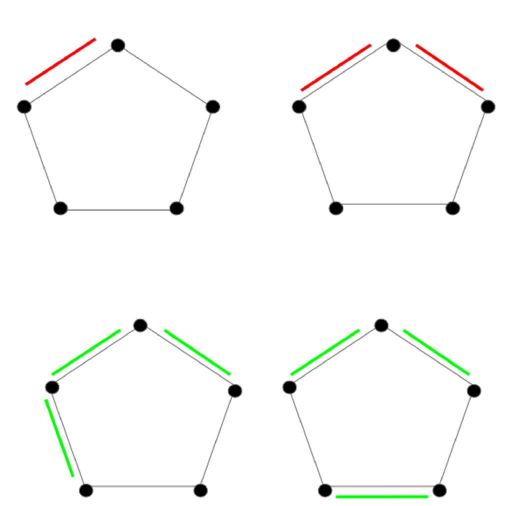


Figure: C_5 is 3-edge dense and not 1- or 2-edge dense.

Objective

Our objective for the summer was to find the minimum edge densities for as many different graphs as we can. Our results showcase the patterns we have noticed and expanded upon while finding minimum edge densities.

Propositions

Proposition 1: If G is a graph with n-many edges, then G is n-edge dense.

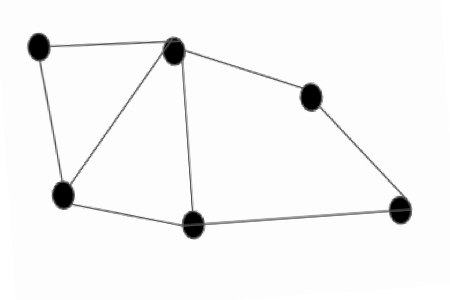


Figure: A graph with 8 edges is 8-edge dense.

Proposition 2: If G is d-edge dense, then G is $(d+1)$-edge dense.

Proof. For something to be d-edge dense, then all possible combinations of subgraphs that have d amount of edges are either edge blankets or have internal edges. Looking for $d+1$ is unnecessary since then you would be adding edges to possible subgraphs that already are edge blankets or have an internal edge. □

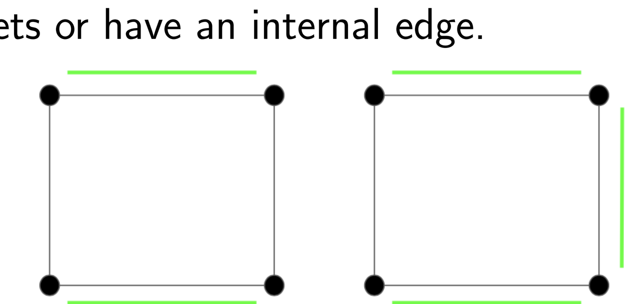


Figure: C_4 is 2-edge dense and thus 3-edge dense.

Star Graph: A graph G is a *star graph* if there exists a $v \in V(G)$ such that for all $e \in E(G)$, we have $v \in V(e)$.

Proposition 3: For any star graph G, its minimal d-edge density is one.

Proof. We take any edge $e \in E(G)$. This e touches the center vertex because all edges on a star graph touch the center vertex. Since all other edges touch the center vertex, then e is an edge blanket. □

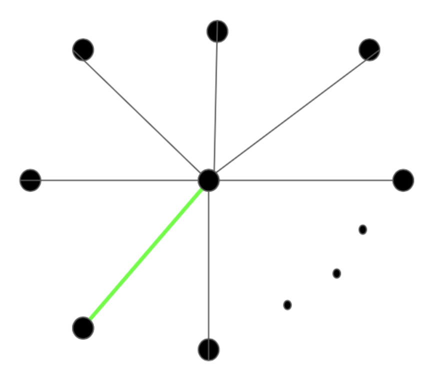


Figure: One edge of S_n is always an edge blanket.

Conjectures

Conjecture 1: Let K_n be the denote the complete graph on $n \geq 3$ vertices. The minimal edge density for K_n is $n - 2$.

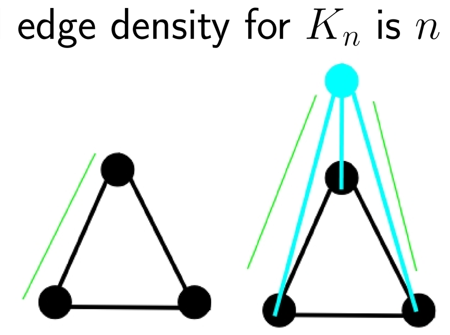


Figure: K_3 with 1-edge density becoming K_4 with 2-edge density.

Remark. K_1 is a particularly special case. Notice that because K_1 has no edges, it is vacuously true that K_1 is d-edge dense for every positive integer d. We encourage the reader to ponder K_2 analogously.

Cycle Graphs: Let G be a single component graph with vertex set $V(G)$ and edge set $E(G)$. G is a *cycle graph* if every $v \in V(G)$ has a degree of 2.

Conjecture 2: If C_n is the cycle graph on n vertices, then the minimal edge density is:

$$\begin{cases} \frac{2}{3}n - 1 & n = 3k, \quad k \in \mathbb{N} \\ \frac{2}{3}n - \frac{1}{3} & n = 3k - 1, \quad k \in \mathbb{N} \\ \frac{2}{3}n - \frac{2}{3} & n = 3k + 1, \quad k \in \mathbb{N} \end{cases}$$

The first few minimal edge densities for C_n that we computed by hand are shown below:

vertices	2	3	4	5	6	7	8	9	10	11	12	13
edge density	1	1	2	3	3	4	5	5	6	7	7	8

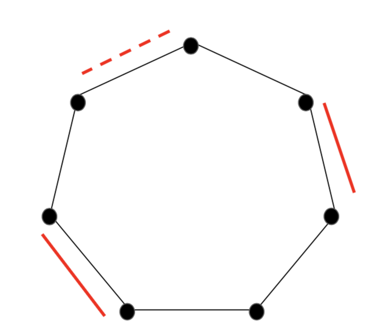


Figure: C_7 is *not* 3-edge dense.

Proof Strategy: Use induction to show:

- The graph is not $(d-1)$-edge dense.
- All subgraphs with d many edges are an edge blanket or have an internal edge.

Theorem

Flower Graph: A *flower* is a graph where all the petals share a vertex in the center of the graph. A petal consists of three edges and three vertices, including the shared vertex (e.g. a triangle). Two edges of a petal are connected to the center vertex and the third edge is connected between the two non-center vertices.

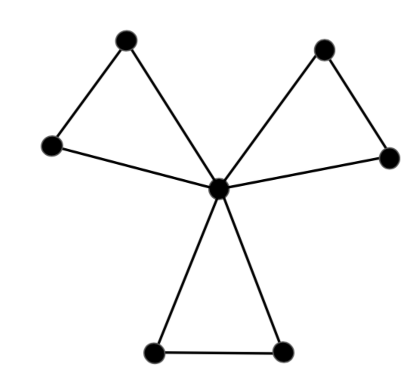


Figure: The flower with three petals, F_3.

Theorem. The minimal edge density of every flower graph F_k with k-many petals is $2k - 1$. Hence, all flowers have an odd minimal edge density.

Proof. Because we can build F_{k+1} from F_k by adding two vertices and three edges, induction is a great way to prove this. The base case for F_1 was verified when creating the table in Conjecture 1. Assume F_k has a minimum edge density of $2k - 1$ for some $k > 1$. We will show that F_{k+1} has a minimum edge density of $2(k+1) - 1 = 2k + 1$. That is, we argue that an arbitrary subgraph of F_{k+1} with $2k + 1$ many vertices is either an edge blanket or contains an internal edge.

Notice that every subgraph of F_{k+1} with $2k+1$ many edges can be built from a subgraph of F_k with at least $2k - 2$ many edges. Hence, we have 3 cases to discuss:

- When the arbitrary subgraph contains two edges in the new petal,
- when the arbitrary subgraph contains one edge in F_k and one edge in the new petal, and
- when the arbitrary subgraph has no edges in the new petal.

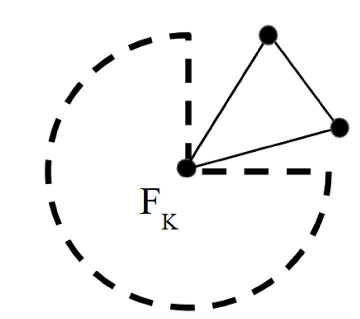


Figure: F_{k+1} built from F_k.

Case 1 and 2: For the first 2 cases, if F_k already has an internal edge then it doesn't really matter. But if it's an edge blanket, putting one or two new edges on the new petal completes the edge blanket again.

Case 3: For the third case, we're proving that putting two new edges in F_k will create an internal edge, given that it already doesn't have one. The maximum number of edges we can have on F_k without having 3 edges on a petal is $2k$. However, when we add the two edges into F_k there will be $2k + 1$ many edges, giving us a 3-cycle and thus an internal edge.

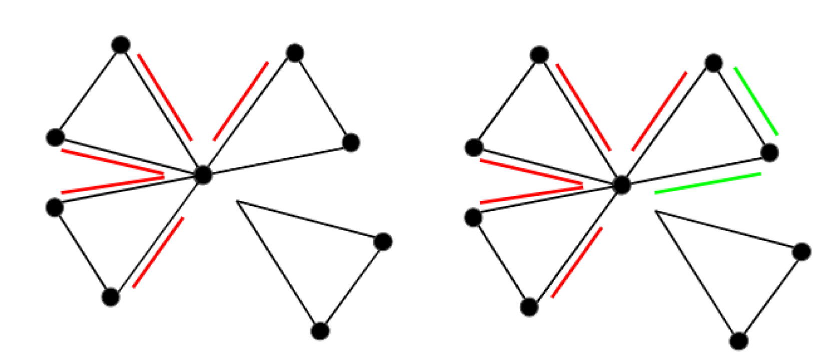


Figure: A petal being added onto a F_3 graph.

One can also use induction to show that $2k - 1$ is indeed the *smallest* edge density of F_k. □

Acknowledgments

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